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1 Hands-on for 2D iterative reconstructions of simulated data with the CASToR platform

1.1 0. Initializations

Just run it as it is

1.2 1. Definition and generation of the phantom

Background disk with smaller contrast disks

Generate both an ASCII file describing the phantom and a pixelized image of the phantom

Specify:

- the radius of the background disk in mm (radius), default = 100
- the emission density of the contrast disks relative to the background (sphereValues), default = 1

1.3 2. Simulation and reconstruction of a noise-free sinogram

1.4 3. Simulation and reconstruction of noisy data

Select the number of events (eventsNb) such as to get a reasonably noisy image

- Optimization algorithm: MLEM, 14 subsets, 4 iterations (default values in clinics)

1.5 4. Effect of the number of iterations for the ML-EM algorithm for noisy data

For the number of events selected above, the 45 slices are reconstructed independently in 2D. Each slice can therefore be considered as independent. The mean and the standard-deviation images are computed from the 45 slices.

Qualitative assessment: evaluate visually the quality of the images (single slice, mean and standard-deviation images) as a function of the number of iterations.

Quantitative assessment: evaluation of the signal versus noise trade-offs as a function of the number of iterations. Discuss the trends of the curves and the differences in convergence between positive contrast (activity > background) and negative contrast (activity < background).

Figures of merit for the signal and the noise: * Signal: Contrast Recovery Coefficient (CRC) of the contrast disks, measured on the mean image. Contrast is defined as the ratio between (diskActivity

- backgroundActivity) and (backgroundActivity). The recovery coefficient is the ratio between the measured contrast and the nominal contrast.

* Noise: Mean value in the center of the background, measured in the standard-deviation image.

Reconstruction default parameters: * MLEM: opti='MLEM' * Number of iterations: iterNb=24 * Number of subsets (OSEM): subsetNb=7

1.6 5. Effect of the prior and the hyper-parameter value (beta) for the MAP algorithm for noisy data

Evaluation of MAP reconstructions for the three different potential functions for the Markov random field prior (page 140): * Quadratic * Huber * Relative difference

For the number of events selected above, the 45 slices are reconstructed independently in 2D. Each slice can therefore be considered as independent. The mean and the standard-deviation images are computed from the 45 slices.

Qualitative assesment: evaluate visually the quality of the images (single slice, mean and standard-deviation images) as a function of the value of the hyper-parameter beta and the potential function

Quantitative assesment: evaluation of the signal versus noise trade-offs as a function of the hyper-parameter beta. If beta equals zero, what does it mean?

Discuss the differences between MLEM and MAP

Discuss the differences between the different priors (potential functions)

Figures of merit for the signal and the noise: * Signal: Contrast Recovery Coefficient (CRC) of the contrast disks, measured on the mean image. Contrast is defined as the ratio between (diskActivity - backgroundActivity) and (backgroundActivity). The recovery coefficient is the ratio between the measured contrast and the nominal contrast.

* Noise: Mean value in the center of the background, measured in the standard-deviation image.

Reconstruction default parameters: * MLEM: opti='PPGML' * Number of iterations: iterNb=24 * Number of subsets (OSEM): subsetNb=7 * Values of the hyper-parameter, beta. It depends on the number of events. It ranges from 0 to betaMax such as the noise at betaMax is similar to the noise with ML-EM after one iteration. Example of values: betas=[0,0.125,0.5,2.0,9.0]. Always use the same number of beta values. * penalty: 'quadratic' or 'huber' or 'nuyts' * Anatomical prior: bowsher=None

1.6.1 Discuss the effect of an anatomical prior

Reconstruction with matched anatomical prior: * MLEM: opti='PPGML' * Number of iterations: iterNb=24 * Number of subsets (OSEM): subsetNb=7 * Values of the hyper-parameter, beta. It depends on the number of events. It ranges from 0 to betaMax such as the noise at betaMax is similar to the noise with ML-EM after one iteration. Example of values: betas=[0,0.125,0.5,2.0,9.0]. Always use the same number of beta values. * penalty: 'quadratic' * Anatomical prior: bowsher=1

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